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(1) Research Program/Project/Study Title:

Development of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*)

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Program or Project has component (if no proceed to 2)? ☐ Yes ☐ No

Component 1 Title:	
Component Leader/s:	
Academic Rank:	

(2) Category (please check)

- ☐ Social Science/Education /Humanities & other related fields
☐ Natural/Biological Sciences
 ▪ Food Research
☐ Developmental/ Computer/Engineering/ Architecture
☐ Extension

(3) Research Center

[Check the concerned center]

- ☐ SSITRDC
 Social Science and Industrial Technology Research Development Center

☐ LRDC
 Livestock Research Development Center

 ▪ **FFRDC**
Fisheries and Food Research Development Center

☐ CSRDC
 Crop Science Research Development Center

(4) Site of implementation:

Municipality/City:	Roxas City
Province:	Capiz
Region:	VI

(5) Classification

- Research:** **Development:**
 ▪ Basic ☐ Pilot Testing
☐ Applied ☐ Tech. Promotion/Commercialization

(6) Mode of Implementation

- ☐ Single Authorship/Proponent
 ▪ Collaborative Research (within campus)
☐ Collaborative Research (intercampus)

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	☐ Collaborative Research (external agency)
(7) Sector/Commodity NUTRITIOUS JUICE	(8) Discipline/Field of Specialization Food and Technology

(9) Rationale

A small citrus fruit called calamansi, "Citrus microcarpa", often called calamondin or Philippine lime, is widely grown in the Philippines and other Southeast Asian nations. Due to its sour and acidic flavor, it is a common component in Filipino and different Southeast Asian cuisines.

Calamansi is a versatile fruit valued for its unique flavor and versatility in cooking. Its use extends beyond the kitchen into various cultural, culinary, and practical applications.

Typically, calamansi fruits are small, with diameters of 2-4 centimeters. They have a vivid orange or yellow-orange color and resemble little oranges.

Calamansi is a good source of vitamin C, which is necessary for a healthy immune system and skin, just like other citrus fruits. Antioxidants and other beneficial ingredients are also present. Calamansi has been used in traditional medicine for some ailments, including coughs, colds, and sore throats. The immune system is supposed to benefit from its high vitamin C concentration.

Cucumbers, related to melons, such as watermelon, cantaloupe, and honeydew, are a relatively low-calorie food at just about 15 calories per cup and are about 95% water. They contain high levels of lignans, vitamin K, cucurbitacins and their derivatives (triterpenoids), flavonoids (apigenin, luteolin, quercetin, and kaempferol), antioxidants such as beta carotene and vitamin C, and B vitamins, among other trace elements and minerals. With such a high level of water content and the bonus of naturally occurring nutrients and trace minerals, cucumbers could be excellent supplements to drinking water or even serve as an alternative to consuming sports drinks. (H. Murad, M.A. NYC)

Ginger (*Zingiber officinale*) is one of the most commonly consumed dietary condiments in the world (Surh et al. 1999). The oleoresin (i.e., oily resin) from the rhizomes (i.e., roots) of ginger contains many bioactive components, such as [6]-gingerol which is the primary pungent ingredient that is believed to exert a variety of remarkable pharmacological and physiological activities. Ginger has been used for thousands of years for the treatment of numerous ailments, such as colds, nausea, arthritis, migraines, and hypertension.

This study aims to provide a nutritious juice utilized by Calamansi with cucumber and ginger to create an additional product, extend its shelf life, and increase the production of Calamansi, Cucumber and Ginger in the province of Capiz.

(9-A) Project Description *(Discuss the project strategy and plan of action of the project. It should also describe the technical procedures and other work needed to accomplish the project objectives. The relationship of the project inputs and activities should be explained clearly in this section. Projects similar/related to the proposed project should be mentioned.)*

The following ingredients comprise the nutritious juice that will be evaluated and analyzed by the research: the primary ingredient, calamansi juice, which is abundant in nutritional elements and readily available locally: ginger and cucumber, water, agar, and refined sugar. The Nutritious Juice's acceptance will be determined by its sensory attributes, including color, texture, taste, and odor. Additionally, the acceptability of Nutritious Juice by consumers will be assessed.

This five-month project will involve material preparation, product development and evaluation, statistical analysis, consumer acceptability, and write-ups at the FFRDC laboratory.

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(10) Objectives/Statement of the Problem

The aim of the study is to develop a juice concentrate from calamansi, Cucumber and Ginger.

This study aims to,

1. To determine the acceptability of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*).
2. To evaluate the sensory properties of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*).
3. To assess consumer acceptability of the optimized product;
4. To determine the self-life of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*).

(11) Review of Literature/Prior Art Search (Related researchers/activities that have been conducted. State-of-the-art of current technology/ information from which the project proposal will take off).

Intervention of calamansi juice [*Citrus microcarpa*] to improve lipid profile of overweight adult women. The objective of this research was to analyze the effect of calamansi juice intervention on lipid profiles of overweight adult women. The pre-post and randomized single-blinded clinical study were applied on 24 subjects who were randomly and allocated into two group, i.e group A (300 ml 100% pure calamansi juice) and group B (300 ml and 50% concentrante calamansi juice). The juice was given twice a day (each 150 ml) for 28 days and lipid profiles were analyzed before and after intervention. The result showed a significant decrease ($p < 0.05$) after intervention in Triglyseride plasma (TG) level (-23.67 ± 21.81 and -2.33 ± 22.39 mg/dl), low density lipoprotein (LDL-c) -30.58 ± 25.05 and 12.42 ± 26.66 mg/d), total cholesterol (TC) 23.50 ± 30.90 and 12.00 ± 30.80 mg/dl and a significant increase ($p < 0.05$) in high density lipoprotein (HDL-c) plasma level (10.25 ± 11.91 and 0.33 ± 8.57 mg/dl). Meanwhile, in group B there was no significant change in lipid profiles. The intervention of 300ml pure calamansi juice improved the lipid profile of overweight women. (Sutemy Surlitah, Budi Setiawan, Dodik Briawan DOI: 10.25182/jgp.2017.12.2.93-100)

Contribution of Grape Juice to Develop New Isotonic Drinks with Antioxidant Capacity and Interesting Sensory Properties. This research aims to develop a juice concentrate from Honey, ginger and calamansi. Specifically, it aims to: determine the most accepted formulation of the juice concentrate and determine sensory characteristic of the juice concentrate in terms of color, taste and aroma. Based on the results the most accepted formulation of the juice concentrate is the first formulation, a blend of 50% of the calamansi concentrate and ginger concentrate with 40% Honey. Sensory characteristic of the juice concentrate in terms of color is yellow-orange, taste is Spicy-warm, zesty, sweet, sour with a bit earthy taste and aroma is Spicy-warm, zesty, sweet, sour with a bit earthy taste. It is further recommended to test Physico – Chemical, Microbiological and Shelf life testing of the juice concentrate which indicates that the product is safe to consume. (Bendaali et al., 2022)

PH22018000251U1 Lemon Grass Beverage Enhanced with Calamansi

A natural and nutritious health drink particularly a lemongrass juice enhanced with calamansi comprising the following ingredients: lemongrass juice- 240ml; calamansi juice- 20 grams; and syrup - 20 grams.

CN101999720A Method for processing high-pulp content and concentrated lime juice.

The invention belongs to the field of orange juice processing and particularly relates to a method for processing high-pulp content and concentrated lime juice. The method comprises the following steps of: 1) soaking lime fruits into potassium permanganate solution for treatment, spraying chloridized water onto the surfaces of the fruits for sterilization, and spraying the fruits

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with clean waterto obtain clean fruits; 2) squeezing the fruits by using a centrifugal juicer, and collecting lime juice and peel residues; 3) slicing the lime peel residues; 4) debitterizing the lime juice by using macroporous resin HDP800, and debitterizing lime peel in a water soaking mode; 5) adding sugar into the lime peel residues, and curing, freezing and storing the lime peel residues; 6) adding honey into the frozen and stored lime peel residues, concentrating the lime juice, vitamin C, citric acid, goldenpomelo essence and the like until the heating temperature is 100 DEG C to obtain semi-finished products of the concentrated lime juice; and 7) sterilizing the lime juice at the temperature of 85 DEG C for 15 minutes to obtain a high-pulp content and concentrated lime juice product. By the method, the flavor substances in the high-pulp content lime juice are preserved to the greatest extent. The method has important significance to the improvement on added value of quince processing industry.

CN103519296A Processing method of clear cucumber juice with good taste.

The invention provides a processing method of clear cucumber juice with good taste, which comprises the following steps of (1) processing raw materials; (2) separately processing the peel, pulp, base and flesh into concentrated juice, and producing concentrated juice by use of the flesh; (3) enzyme deactivation: performing sterilization and enzyme deactivation on the syrup for 30 seconds at a temperature over 95 DEG C, and then cooling the syrup to 50-55 DEG C; (4) ultrasonic enzymolysis: performing enzymolysis on the syrup under an ultrasonic condition, adding amylase and pectinase into the syrup, performing enzymolysis for 0.5-2 hours, and then adding inscribed and circumscribed protease to finally decompose protein into amino acid small molecules, thereby improving the juice yield. In the method, enzymolysis is performed twice, and the cucumber juice is hydrolyzed by use of the amylase, pectinase and protease, so that nutrient substances in the cucumber juice are hydrolyzed into nutritional components which are easy to absorb by human bodies and have special physiological functions; moreover, by adopting the processes such as adsorption, ultrafiltration and the like, the final product of concentrated clear cucumber juice has high nutritional value, stable characters and good flavor and taste.

Ph22018000725u1 formulation of making calamansi juice with mango pulp.

This utility model relates generally to the food industry particularly to the formulation of fresh nutritious and refreshing drink. These six (6) ingredients of calamansi juice comprising of fresh calamansi juice, blended ripe mango pulp, white sugar, honey, purified boiling water and pinch of salt.

(12) Scientific Basis/Theoretical Framework (Include concepts. theories and variables that indicate relevant relationships to be tested.

INPUT	PROCESS	OUTPUT
Calamansi Juice and other ingredients	- Product Preparation and formulation of Calamansi Juice with Cucumber and Ginger - Consumer Testing and sensory evaluation - Shelf-life testing	- Optimized formulation of Calamansi Juice with Cucumber and Ginger - Shelf-stable product - Proximate Analysis of best formulation

(13) Methodology (Details such as; a) variables to be measured and evaluated; b) treatments to be used and their layo c) experimental procedures and design; d) statistical analysis; e) evaluation method and observations to be made, strategies for implementation.)

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The study will be experimental research. It will focus on Development of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*) and conduct the exact proportion

Table 1. Treatments of Calamansi, Cucumber and Ginger Juice

Ingredients	T1	T2	T3	T4
Calamansi Juice	100ml	100ml	100ml	100ml
Water	500ml	500ml	500ml	500ml
Cucumber	100g	95g	90g	85g
Ginger	0	5g	10g	15g
Diluted Refined Sugar	140g	140g	140g	140g
Agar	5g	5g	5g	5g

Utensils:

- Knife
- Chopping Board
- Casserole
- Bowl
- Stove
- Steamer
- Ladle
- Measuring cup and Spoon
- Strainer
- Blender
- Plastic Bottles

Preparation for Plastic Bottles

1. Washing the Bottles and Cups
2. Sterilizing the Bottles and Cups

Preparation for Diluting Refine Sugar

1. Dissolve 150g sugar in 80 ml water over low heat.

Procedures for Making the Juice

1. Preparing all the ingredients
2. Washing the Calamansi, Cucumber and Ginger
3. Squeezing the Calamansi using a citrus squeezer or you can use your hands
4. Straining the squeezed Calamansi Juice to separate the seed from the Juice.
5. Putting all the ingredients together in a blender (Cucumber, Ginger, Agar, Diluted Refined Sugar, Calamansi Juice, and Water)
6. Blending all the Ingredients together.
7. Straining out the mixed Juice to separate the pulp.
8. Sterilizing by boiling the Juice over a low-heat fire.
9. Cooling the Juice
10. Transferring the juice into a sealed container and refrigerate.

Experimental Design

This study will use the experimental method or procedure involving the control or manipulation of conditions to study the relative effects of various treatments applied to several samples or the same treatment applied to several samples. A completely randomized block design (CRD) will be employed during the treatment and sensory evaluation.

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Proximate analysis

The Calamansi Juice with Cucumber and Ginger will be brought to the DOST laboratory for proximate composition analysis.

Statistical analysis

The data collected on sensory evaluation was statically analyzed by using SPSS software. The analysis of variance (ANOVA) was used to analyze sensory evaluation results with the significant difference among the samples with different treatments.

Data Collection

The data was gathered using a score containing the 9-point hedonic scale. It includes the panelist's color, taste, flavor, texture, and general acceptability rating. The data was gathered after every replication and was collected, tallied, tabulated, and computed using mean for acceptability and Analysis of Variance (ANOVA) for the significant difference among treatments.

Sensory evaluation

The Calamansi Juice with Cucumber and Ginger will be prepared per the sensory evaluation protocol.

Scorecards that contain the 5-point scale for sensory characteristics and the 9-point Hedonic Scale for acceptability will be used to evaluate the development of the product, and 30 evaluators will assess the product.

Only sensory characteristics such as color, taste, flavor, texture, and general acceptability will be evaluated.

For the consumer's preference, 100 respondents of varying ages and sex will evaluate the product.

Scoring Variables

The Criteria for evaluation of the product's sensory characteristics used a 5-point hedonic scale in terms of color, taste, flavor, texture, and general acceptability. The rating for general acceptability in 9- point hedonic scale is 9 for like extremely, 8 for like very much, 7 for like moderately, 6 for like slightly, 5 for neither like nor dislike, 4 for dislike moderately, 3 for dislike slightly, 2 for dislike very much and 1 for dislike extremely (Calmorin 2006).

The score on each item will be interpreted based on the result of the computation regarding color, taste, flavor, texture, and general acceptability using the following scoring interval and verbal interpretation.

General Acceptability

Scoring	Score Interval	Verbal Interpretation/Response
9	8.12-9.0	Like Extremely
8	7.23-8.11	Like Very Much
7	6.34-7.22	Like Moderately
6	5.45-6.33	Like Slightly
5	4.56-5.44	Neither Like nor Dislike
4	3.67-4.55	Dislike Slightly
3	2.78-3.66	Dislike Moderately
2	1.89-2.77	Dislike Very Much
1	1.00-1.88	Dislike Extremely

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(14) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Research Objectives	Duration (months)				
	1	2	3	4	5
1. Preparation and selection of Raw Materials					
2. Product Development					
3. Product Evaluation					
4. Shelf life Testing					
5. Statistical Analyses					
6. Manuscript preparation and write-up					

(15) Expected Output (Indicate the specific products, processes or services which the project aims expected to produce; quantify when possible)

1. Utility Model
2. Patent Product
3. IEC material
4. Commercialization Product

(16) Technology Transfer Activity Design (Indicate the specific products, activity design, processes or services to be transferred as extension activities)

Training on the process of Calamansi (*Citrus microcarpa*) Juice with Cucumber (*Cucumis sativus*) and Ginger (*Zingiber officinale*)

(16) Targets Beneficiaries (Who the clientele are and what are the expected outcome/effects of the use of the projects output)

1. Calamansi, Cucumber and Ginger farmers
2. Entrepreneurs
3. Researchers
4. Consumers
5. Vegetarians

(17) Personnel Requirement

Name	Role (Program, Project, Study Leader)	Field of Specialization
Stephanie S. Pimentel	Study Leader	Biology
Janine G. Ferrer	Research Assistant	Biology
Vialyn B. Ragalado	Research Assistant	Food Science

Percent time devoted to the Program/ Project/ Study

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Program/Project Leader (in hours):		
Program/Project Members (in hours):		
(18) Budgetary Requirement		
Ingredients	Quantity	Amount
Calamansi	50 kls	1,200.00
Other Ingredients (Cucumber, Ginger, Refined Sugar, Water)		1,100.00
Gas during product Development.		1300.00
Plastic cups for Sensory and Plastic Gloves	6packs	1000.00
Proximate Analysis		10,000.00
Transportation Allowance		1000.00
Patent		5,000.00
TOTAL		20,600.00
(19) Literature Cited (include only those references published in reputable journals, and at least that from CAPSU Research Journal, http://researchjournal.capsu.edu.ph/) Albaladejo Dr Reyma P [Ph], Univ Capiz State [Ph], Ph22018000251u1·2018-08-22. Lemon Grass Beverage Enhanced With Calamansi		
(20) Capsule Curriculum Vitae (One-page brief C.V. only)		